Range Hood Design Guide

A Field Reference for Kitchen Designers

Introduction

Range hoods are one of the most technically exacting elements in kitchen design — and one of the most commonly specified incorrectly. A hood that is undersized, hung too high, or poorly positioned will underperform, create noise problems, or require expensive rework. This guide covers everything you need to know to specify, position, and design around range hoods with confidence.

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1. Core Principle: Capture Is Everything

The job of a range hood is to capture grease, steam, smoke, and odors *before* they escape into the kitchen. Every sizing and placement decision you make flows from this one goal. A hood that looks beautiful but allows a plume of smoke to escape to the sides is a failed specification.

Think of the rising column of heat and steam from a cooktop as a cone that expands as it rises. Your hood needs to intercept that cone before it spreads beyond the hood's perimeter.

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2. Hood Width: Sizing Relative to the Cooktop or Range

The Baseline Rule

The hood should be at least as wide as the cooktop or range below it. This is the absolute minimum — anything narrower is a compromised specification.

The Professional Standard

Add 3 inches on each side (6 inches total) beyond the cooktop width. This overhang captures the lateral drift of the thermal plume, especially at higher BTU outputs.

Cooktop / Range Width	Minimum Hood Width	Recommended Hood Width
24"	24"	30"
30"	30"	36"
36"	36"	42"

48"	48"	54"–60"
60"	60"	66"–72"

High-BTU and Professional-Grade Ranges

For commercial-style or pro ranges (Wolf, BlueStar, Thermador, Bertazzoni, ILVE, etc.) outputting 15,000–25,000+ BTU per burner, **size up aggressively**. A 36" pro range should minimally have a 42" hood, and ideally a 48" hood. The lateral plume spread at high heat is significantly greater than on a standard residential cooktop.

Wall Hoods vs. Island Hoods

- **Wall hoods** need the 3" per side overhang as described above.
- **Island hoods** are exposed on all four sides and should overhang **6 inches on all four sides** — not just front-to-back but also left-to-right. An island cooktop positioned off-center from the hood is one of the most common design mistakes you'll encounter.

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3. Hood Depth: Don't Neglect the Front-to-Back Dimension

Hood depth is undersized almost as frequently as width.

Wall Hoods Over Ranges

Standard residential ranges are 24"–25" deep. Your hood should cover the full depth of the cooking surface. Most standard wall hoods are 20"–24" deep — this is typically adequate for a standard range but worth confirming for deeper commercial-style units.

Island Hoods Over Cooktops

Island cooktops vary in depth from 18"–21". The hood must cover the full cooktop depth with ideally a 3"–6" overhang on the user's side (the front) to capture steam rising toward the cook.

Key check: Confirm cooktop depth against hood depth at specification time. Don't assume.

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4. Mounting Height: The Most Critical Dimension

Standard Height Ranges

The vertical distance is measured **from the cooking surface (burner grate top) to the bottom of the hood**.

Hood Type	Minimum Height	Maximum Height	Optimal Range
Standard residential hood	18"	30"	24"–30"
Pro / high-BTU range hood	24"	36"	27"–36"
Downdraft (pop-up)	N/A – surface-mounted	–	–

Why Height Matters

- **Too low (under 18"):** Creates a burn/fire hazard. The hood itself can overheat, and the operator's head enters the clearance zone. Manufacturer minimum clearances must be respected – always check the spec sheet.
- **Too high (over 36"):** The thermal plume expands too wide before reaching the hood. Capture efficiency drops dramatically. The hood must work harder (higher CFM required) to compensate.
- **Sweet spot (24"–30" for residential):** Captures the plume before significant lateral spread, keeps the hood within comfortable sight lines, and maintains code-compliant clearances from the cooking surface.

Critical Rule: Always Check Manufacturer Minimums

Every hood has a listed minimum clearance from the cooking surface, typically 18"–24" for electric/induction and 24"–30" for gas. **These are code-required fire safety minimums, not design preferences.** Never spec below them.

Cabinet Height Considerations

The most common conflict in kitchen design is fitting a hood between the cooktop and an upper cabinet or soffit above. Here's the math:

- Standard floor-to-ceiling: 96" (8 ft)
- Standard base cabinet + countertop + range: 36"
- Desired hood bottom: 36" + 24"–30" = 60"–66" AFF (above finished floor)
- If upper cabinets end at 84" AFF, hood height available = 84" – 60" = 24" tall hood (minimum)
- If upper cabinets end at 96" AFF, hood height available = 96" – 60" = 36" tall hood

Run this math for every kitchen. In rooms with 9' or 10' ceilings, you often have more flexibility; in rooms with 8' ceilings, you may need to select a hood with a shorter body or a flush/integrated style.

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5. CFM (Airflow): Sizing the Fan to the Cooking Equipment

The NKBA / HVI Standard

The Home Ventilation Institute (HVI) recommends a minimum of **1 CFM per 100 BTU** of total cooktop output.

Example: A 4-burner range with 4 × 12,000 BTU burners = 48,000 BTU total → minimum 480 CFM.

A simpler residential rule of thumb: **100 CFM per linear foot of hood width.**

Hood Width	Minimum CFM (Residential)	Recommended CFM (Pro Range)
30"	300 CFM	500–600 CFM
36"	360 CFM	600–900 CFM
42"	420 CFM	900–1,200 CFM
48"	480 CFM	1,000–1,500 CFM

Makeup Air

This is the single most overlooked issue in range hood specification, especially on high-end projects.

Any hood rated at 400 CFM or more will depressurize the kitchen if makeup air is not provided. At 600+ CFM, doors can be difficult to open, gas appliance back-drafting becomes a hazard, and HVAC systems may malfunction.

Your specification should always include:

- A note on whether the kitchen requires a makeup air system
- Coordination with the HVAC contractor for homes with tight building envelopes
- Many jurisdictions now require makeup air for hoods over 400 CFM by code

Recirculating (Ductless) vs. Ducted

Always specify ducted where possible. Ductless/recirculating hoods filter air through charcoal and return it to the room — they do not remove heat, humidity, or all odors. They are a compromise for situations where duct runs are not feasible (apartments, island installations where ceiling duct routing is cost-prohibitive).

If a client insists on ductless, set expectations clearly: performance will be significantly inferior to a ducted system.

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6. Duct Sizing and Routing

A correctly sized and routed duct is as important as the hood itself. A powerful blower pushing against an undersized or kinked duct is loud and inefficient.

Duct Diameter Minimums

Hood CFM	Minimum Round Duct Diameter	Minimum Rectangular Duct Equivalent
Up to 400 CFM	6" round	3.25" × 10"
400–600 CFM	7"–8" round	3.25" × 14"
600–900 CFM	8"–10" round	6" × 10"
900–1,200 CFM	10"–12" round	8" × 12"
1,200+ CFM	12"+ round	Consult manufacturer

Routing Best Practices

- **Shorter is better.** Each additional foot of duct run reduces effective CFM.
- **Every 90° elbow = approximately 10 feet of equivalent duct length.** Minimize elbows; use 45° bends where possible.
- **Total effective duct length should not exceed 30–40 feet** for residential applications (consult hood manufacturer for their specific limits).
- **Always terminate outside** — never into an attic, crawlspace, or wall cavity.
- Use a **backdraft damper** at the exterior termination point.
- **Insulate duct runs** that pass through unconditioned spaces to prevent condensation.

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7. Hood Types and Design Contexts

Wall-Mount Chimney Hoods

The most common type. Mount on the wall above a range or wall cooktop. Available in 30"–66" widths. The chimney portion extends to the ceiling and conceals the duct. Works best when the cooktop is on an exterior or interior wall with a clear path to the roof or soffit for duct routing.

Design note: The chimney section is sized to reach the ceiling. Measure from the top of the hood body to the ceiling and confirm the chimney extension kit covers that

distance. Most hoods come with extensions for 9' and 10' ceilings; custom orders may be required for 11'+ ceilings.

Under-Cabinet Hoods

Installs beneath an upper cabinet directly above the cooktop. Generally limited to 300–600 CFM. Ideal for standard kitchens where cabinetry runs above the range location. Not appropriate for high-BTU pro ranges.

Design note: Confirm the upper cabinet's bottom shelf height against the hood's mounting height requirement. The hood must be accessible for filter cleaning.

Insert / Liner Hoods

A blower insert housed inside a custom or semi-custom hood surround — typically a furniture-grade wood or plaster structure built by a cabinetmaker. Common in transitional and traditional kitchens where a visible appliance aesthetic is undesirable.

Design note: The interior dimensions of the surround must accommodate the insert's airflow requirements. Work with the insert manufacturer's specified clearances. Do not reduce the opening size for aesthetic reasons — you will choke the airflow.

Island Hoods

Suspended from the ceiling above an island cooktop. Requires a ceiling-mounted duct connection, which must be planned in the structural/MEP coordination phase — not after the ceiling is drywalled.

Critical coordination items:

- Ceiling joist or beam location for blocking (hoods can weigh 80–200+ lbs)
- Electrical junction box in ceiling centered on hood location
- Duct chase through ceiling cavity — must be confirmed before construction

Downdraft / Pop-Up Hoods

Rise from behind the cooktop or from within the countertop. Performance is inherently limited because they work against the natural upward movement of heat. Generally appropriate only for light cooking applications on induction or lower-BTU gas where a wall or island hood is not feasible.

Honest guidance for clients: Set expectations. Downdrafts are a design compromise. They are better than nothing for an induction cooktop used for everyday cooking, but they will not satisfy a client who cooks aggressively.

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8. Clearance and Code Considerations

Always verify against local codes (IRC, local amendments) and the specific hood and range manufacturer's installation instructions. These are your primary references.

Clearance Item	Typical Requirement
Hood bottom to gas cooktop surface	24"–30" minimum (check manufacturer)
Hood bottom to electric/induction cooktop	18"–24" minimum (check manufacturer)
Hood to combustible side wall (if applicable)	Per manufacturer spec, typically 0"–3"

Duct exterior termination to window/door	3' minimum (many jurisdictions)
Duct exterior termination to gas meter/AC unit	3'–10' (varies by jurisdiction)

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9. Aesthetic Integration

Centerlines and Symmetry

The hood must be centered on the cooktop, not the range run, not the window above. Draw the centerline from the cooktop and carry it up. If the window above is off-center relative to the cooktop, the hood still centers on the cooktop — the window asymmetry is addressed with surrounding cabinetry design.

Relationship to Upper Cabinets

Wall hoods flanked by upper cabinets should align their bottom with the bottom of adjacent upper cabinet doors. The upper cabinet run typically sits at 54" AFF (bottom of uppers) for an 8' ceiling, 60" AFF for a 9' ceiling. This creates a clean horizontal datum line across the wall.

Proportion to the Wall

A hood that is too shallow in height reads as underpowered and out of scale, even if it performs adequately. As a general proportion guideline, a 30"–36" wide hood should be at minimum 18"–24" tall in its body (not including chimney). Tall-ceiling kitchens warrant taller chimney sections to avoid a floating, disconnected look.

Tile and Backsplash Integration

Plan the backsplash tile layout around the hood before the tile order is finalized. The hood's sides and bottom should return cleanly to the wall surface. Where the backsplash meets the hood sides, a reveal or edge detail should be resolved in the design documents — not left to the installer.

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10. Common Mistakes to Avoid

Mistake	Why It Matters	Correction
Hood same width as cooktop, no overhang	Plume escapes the sides at higher BTUs	Add 3" per side minimum
Island hood not centered on cooktop	Asymmetric capture; looks wrong	Center hood on cooktop centerline
Mounting too high (over 30" residential)	Capture efficiency loss; loud fan required	Bring down to 24"–28" optimal
Undersized duct (4" on a 600 CFM hood)	Noise, backpressure, reduced performance	Size duct to CFM per table above

Forgetting makeup air on 600+ CFM	Depressurization, backdraft risk, code violation	Specify makeup air system
Chimney extension not ordered for ceiling height	Hood won't reach ceiling; looks unfinished	Confirm ceiling height at spec time
Ductless specified on gas pro range	Serious performance gap; client dissatisfied	Always duct pro ranges
No blocking spec'd for island hood	Hood falls; liability issue	Coordinate blocking in structural phase

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11. Specification Checklist

Use this at the time of hood specification on every project:

- [ ] Hood width = cooktop width + 3" each side (or 6" each side for island)
- [ ] Hood depth covers full cooktop depth
- [ ] Mounting height confirmed within manufacturer's min/max and optimal range
- [ ] CFM confirmed against BTU output and hood width formula
- [ ] Makeup air requirement evaluated (400+ CFM → likely required)
- [ ] Duct diameter specified correctly for CFM
- [ ] Duct routing reviewed — elbow count, total equivalent length
- [ ] Exterior termination location confirmed
- [ ] Ceiling height measured → chimney extension ordered if needed
- [ ] Island hood: blocking and ceiling electrical confirmed with contractor
- [ ] Centerline confirmed on drawings — hood centered on cooktop

- [ ] Hood bottom aligned with upper cabinet bottom on flanking walls
- [ ] Backsplash edge condition at hood resolved in drawings
- [ ] All clearance dimensions verified against manufacturer installation manual

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12. Quick Reference Card

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Hood Width          = Cooktop Width + 6" (3" per side)
Island Hood Width   = Cooktop Width + 12" (6" per side)
Mounting Height     = 24"–30" (residential) | 27"–36" (pro/high BTU)
Minimum CFM         = 100 CFM per linear foot of hood width
                      OR 1 CFM per 100 BTU (whichever is greater)
Makeup Air          = Required at 400+ CFM in most jurisdictions
Duct Minimum        = 6" round up to 400 CFM | 8"–10" up to 900 CFM
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Guide prepared for designer training. Always verify manufacturer installation requirements and local building codes on every project. Specifications in this guide represent best practice standards; local codes and manufacturer specifications take precedence where they are more restrictive.